

NASA Turbofan Level B2: Data pretreatment and Modelling Plan

- codes: https://github.com/drvojtex/ADAML_NASATurbofanLevelB2

Data pretreatment

As a first step of data pretreatment, the constant columns (Sensors 1, 5, 6, 10, 16, 18 and 19) were removed. Next, the dataset was randomly shuffled at the unit level and split (also at the unit level) into the three groups. The first group is the training-calibration set (60 % of the units within the dataset), the second group is the validation set (20 % of the units) and the third group is the testing set (20 %).

Next, there was performed a zscore standardization. The mean and standard deviation were computed from the training-calibration set, and these values were used to normalize all three groups. This ensures that the validation and testing data are standardized consistently with the training-calibration data. Standardization removes the effect of different sensor scales so that all variables contribute equally to subsequent analyses (e.g. PCA). Figure 1 shows histograms of all data (time) points after zscore standardization.

Missing values in the dataset were not explicitly addressed, because the dataset does not contain any such records (samples with unknown values of one or more sensors).

Since the sensor time-series of individual units have different lengths, regression will be performed by using a sliding-window approach. For this reason, each time-series was split into overlapping windows, which will be treated as samples. To preserve contextual information, each window (and thus each variable except the target variable) was, as a further pretreatment step, augmented with the slope and intercept of a linear approximation of the preceding window (within the curve). For the first window of each unit, zero context information was used. In this way, each window (except the first one) is not processed as an isolated sequence of points but as a sequence with information about the previous linear trend estimate. Figure 2 illustrates an example using Sensor 3 data. The coloured points represent sensor values within windows, and the dashed line shows the linear context derived from the preceding window.

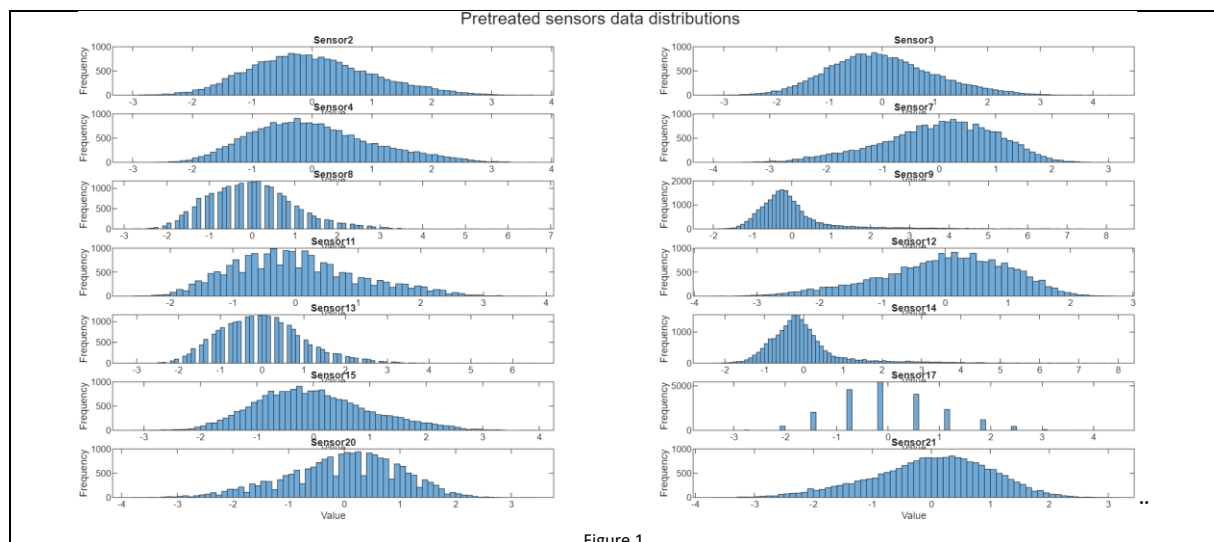


Figure 1

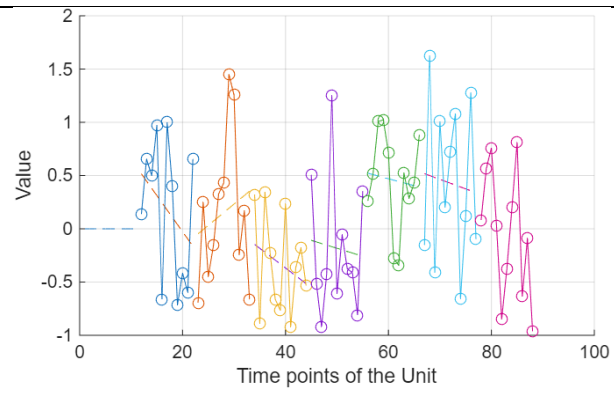


Figure 2

Modelling plan

Modelling goal

The aim is to perform multivariate regression (and evaluate its quality), where the input is a window of measured sensors values (part of the time-series), and the outputs are values of the Sensor 17 of the respective window. Furthermore, the task is also to identify the minimal subset of sensors and the optimal number of latent variables required to predict the target sensor.

Mathematical methods used in modelling

The regression will be performed on windows. Each window of length w will include sensor data and contextual information – intercept and slope of each sensor linear approximation of the previous window (within the curve). The windows will act as samples. Principal Component Analysis (PCA) will be applied to extract latent variables that capture the main sources of variability among the flattened input samples. Principal Component Regression (PCR [1]) will be performed using the most significant principal components to predict the target sensor values (based on the linear least squares). This approach not only reduces the dimensionality of the data but also mitigates multicollinearity among the sensor inputs. The key model hyperparameters are D (set of input sensors), window size w and k (number of PCs); which will be selected based on cross-validation performance. Mathematical formulation of setting PCR parameters is

$$\hat{\beta} = (\mathbf{P}_k^T \mathbf{X}^T \mathbf{X} \mathbf{P}_k)^{-1} \mathbf{P}_k^T \mathbf{X}^T \mathbf{Y},$$

where $\mathbf{P}_k \in \mathbb{R}^{[|D| \cdot (w+2)] \times k}$ is the orthogonal matrix of the first k principal components (each column is a principal component vector, which are eigenvectors of the covariance matrix of $\mathbf{X}$), $\mathbf{X} \in \mathbb{R}^{n \times [|D| \cdot (w+2)]}$ is the matrix of input features and $\mathbf{Y} \in \mathbb{R}^{n \times w}$ is the target matrix. Prediction for the new measurement $\mathbf{x}_{\text{new}}$ is given as $\mathbf{x}_{\text{new}}^T \mathbf{P}_k \hat{\beta}$.

In addition to PCR, Partial Least Squares (PLS [2]) regression will be applied as an alternative latent-variable method. Unlike PCR, which maximizes the variance explained in the input data, PLS directly extracts components that maximize the covariance between predictors and target variables. This makes PLS potentially more efficient in cases where the most informative latent directions are not aligned with the directions of maximal input variance. The mathematical formulation of PLS regression is as follows. The underlying model is

$$\mathbf{X} = \mathbf{T} \mathbf{P}^T + \mathbf{E}, \quad \mathbf{Y} = \mathbf{U} \mathbf{Q}^T + \mathbf{F},$$

where $\mathbf{X} \in \mathbb{R}^{n \times [|D| \cdot (w+2)]}$ is the matrix of input features, $\mathbf{Y} \in \mathbb{R}^{n \times w}$ is the target matrix, $\mathbf{T} \in \mathbb{R}^{n \times k}$ and $\mathbf{U} \in \mathbb{R}^{n \times k}$ are the scores matrices (latent variables extracted from $\mathbf{X}$ and $\mathbf{Y}$), $\mathbf{P} \in \mathbb{R}^{[|D| \cdot (w+2)] \times k}$ and $\mathbf{Q} \in \mathbb{R}^{w \times k}$ are the loading matrices and $\mathbf{E}$ and $\mathbf{F}$ are the residual matrices. The latent components are obtained iteratively by maximizing the covariance between each pair of score vectors (t_i, u_i) :

$$t_i = \mathbf{X} \boldsymbol{\beta}_i, \quad u_i = \mathbf{Y} \boldsymbol{\alpha}_i, \quad \max \text{cov}(t_i, u_i),$$

where the $\boldsymbol{\beta}_i \in \mathbb{R}^{|D| \cdot (w+2)}$ is the weight vector for $\mathbf{X}$ and $\boldsymbol{\alpha}_i \in \mathbb{R}^w$ is the weights vector for $\mathbf{Y}$. The weights matrix is obtained as $\mathbf{B} = [\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_k] \in \mathbb{R}^{[|D| \cdot (w+2)] \times k}$ and the regression coefficient matrix is given as $\hat{\mathbf{B}} = \mathbf{B}(\mathbf{P}^T \mathbf{B})^{-1} \mathbf{Q}^T$. The prediction for a new measurement $\mathbf{x}_{\text{new}}$ is given as $\mathbf{x}_{\text{new}}^T \hat{\mathbf{B}}$.

Model calibration, validation and testing strategies

Model calibration, validation, and testing strategy will be performed using a combination of fixed splits and three-fold cross-validation. The input to the procedure will be clean data – zscore standardized window samples with addition of contextual information. The dataset will be randomly divided (while preserving particular curves – samples coming from the same curve will be in the same part) into five

parts (1–5), with part 4 reserved as a fixed validation set (validation group) and part 5 as a fixed test set (testing group). Three-fold cross-validation is applied to parts 1–3 (training-calibration group) as follows: for each fold, two parts are used to train the PCR model for multiple combinations of hyperparameters (k , w and D), and one part is used as a calibration set to select the hyperparameter (k , w , D) combination with the best performance. After cross-validation, the fixed validation set (part 4) is used to select, among the three folds, the best-performing parameter combination (PCR parameters setup, k , w and D), and the final model is evaluated on the fixed test set (part 5) to provide an objective assessment of performance. R^2 will be computed for training data group, the Q^2 will be computed for validation and testing data groups. The Root mean squared error (RMSE) will be computed for all (training, validation and testing) groups. While R^2 measures how well the model fits the training data, Q^2 evaluates its predictive performance on independent validation or test data using the calibration (training) mean, making it a more reliable indicator of generalization and helping to detect overfitting. Figure 3 shows the strategy.

Next, after determining the optimal window size w and the set of sensors D , the effect of the number of latent variables (k) on R^2 and RMSE will be explored, to identify the dependence of the model complexity on its performance. Furthermore, the autocorrelation of the regression prediction residuals will be analysed, to verify that the residuals behave as white noise and that no systematic temporal patterns remain unmodeled. A comparison of selected time series (predictions versus ground truth) will also be conducted, to visualise the model's ability to reproduce the trend of the missing sensor values based on the observed data. Finally, a biplot will be generated to display the latent variables in relation to the samples, to interpret the latent space and reveal relationships between variables, samples, based on the regression model.

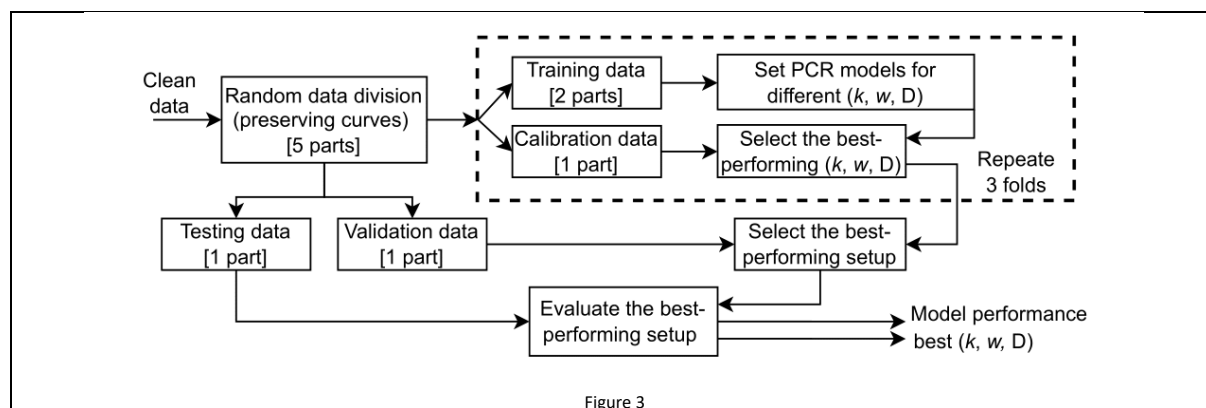


Figure 3

[1] DOI: 10.1002/0471725331

[2] DOI: 10.1007/978-1-62703-059-5_23